

2026

TAKOMO

From atoms to impact. A makerspace for hardware development.

A forge for hardware at Aalto

Aalto Hardware Consortium

WHAT TAKOMO IS

Where ambitious builders come to build the physical future - **together.**

A shared makerspace and community at Aalto, uniting three founding communities - with Aalto University as a key partner.

Thinkin' Rocks

Hardware & compute

Aalto Space Association

Satellites & rocketry

Aalto Robotics Club

Robotics & autonomy

Aalto University

Partner

WHY NOW

“The next big thing is Physical AI - AI with a body. Robots, autonomous machines, industrial systems... It's all coming.”

Jensen Huang — CEO, NVIDIA

We see a shift toward hardware – here are the indicators.

- 1 Capital is moving from models to machines
- 2 Software is becoming abundant
- 3 The industrial base is being rebuilt
- 4 Europe is re-arming and re-industrializing on our doorstep
- 5 **The price is generational. And the consensus is forming.**

Capital, Software, Industry, and Geopolitics are all pointing the same direction: value is moving from software to hardware. The bottleneck is no longer ideas or code – it's funded builders with fast, open place to make things. The talent already exists. The infrastructure is missing.

THE THESIS

As AI makes software cheap, value moves to the physical world - and to the people who can build it.

- 1 Writing software is becoming abundant and near-free.
- 2 What stays scarce: working hardware, embodied systems, and the builders who ship them.
- 3 **The future will be built by people who build. That is us.**

The binding constraint is no longer whether you can code it - it's whether a builder is funded, equipped, and surrounded well enough to make it real.

THE PROBLEM

The problem was never a shortage of ideas. It's the environment to build them.



1

Funding

Hardware costs far more than software - the real crutch.

2

Equipment

Specialised tools, components, and infrastructure.

3

Expertise

Real systems need many disciplines at once.

4

Community

Building goes faster when it happens together.

5

Iteration

Physical systems demand time, testing, repetition.

THE OPPORTUNITY

Finland knows how to build hardware. It always has.

Company	R&D funding	Revenue
Nokia	€4.5B	€19.2B
UPM	€488M	€10.3B
Wärtsilä	€296M	€6.5B
Kone	€203M	€11.1B
Valmet	€123M	€5.4B

Top Finnish R&D spenders, 2024 · Source: Valtioneuvosto

Legacy

A century of building globally recognised technology companies - a foundation for the next generation of deep-tech.

Design

Finnish products win on engineering and world-class design — the combination that competes globally.

THE GAP

Aalto has the talent. It's missing the shared home.

World-class research, engineering talent, entrepreneurial culture, and industry ties.

Yet technical projects still lack one place where researchers, engineers, founders, and industry partners can collaborate, prototype, and turn ideas into real technology.

~150

startups in A Grid (25,000 m²)

Slush

born from Aalto students

**Rovio · Supercell ·
Wolt**

the software lineage

TAKOMO does for atoms what Aalto's students already did for software.

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THE ANSWER

TAKOMO

The forge for Aalto's hardware building.

A shared space, a shared stack of tools, and a shared community where ambitious technical projects get funded, equipped, and shipped. Forged on a day to day basis.

TAKOMO

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HOW IT'S BUILT

One story. Three communities. A clear backbone.



Three proven communities, one roof.

TR

Thinkin' Rocks

Applied engineering lab and prototyping infrastructure for builders, hackers, and founders in hardware and emerging compute.

"Turn white papers into working products."

ASA

Aalto Space Association

Satellite technology, rocketry, and astronomy - behind Aalto-3, a student-built 1U CubeSat, and the EnduroSat rover competition.

Aalto's 6th satellite, student-built.

ARC

Aalto Robotics Club

Real-world robotics: teaching ROS and ML from scratch, competing internationally, building an autonomous logistics robot.

Some identifying fact

TRACTION

We didn't wait for permission. We're already running.

1,000+

event attendees

20+

events (13 YTD)

8

industry partners

3

international collaborations

40+

systems in the hardware library

10+

hardware projects supported

Thinkin' Rocks, last 8 months · plus ASA's student satellites & ARC's international robotics competitions — all before dedicated funding.

THE MODEL

We turn talent into a pipeline that compounds.



↪ people and IP re-enter the TAKOMO stronger each cycle

In: students, researchers, devs, founders - with clarity, passion, direction, and a place to create freely. **Out:** technology and the people who make it.

What a fully-funded day looks like.

Morning

Researchers run a ROS workshop; a founder team mills a housing on the CNC; the hardware library checks out a Jetson and an FPGA board.

Midday

An industry partner drops in with a real-world problem; two members spin up a project around it over lunch.

Afternoon

The satellite team runs a vibration test; a robotics squad iterates on a rover; a solo builder debugs a board beside a PhD who's done it before.

Evening

A build session, a talk, a demo night - and someone decides to turn their prototype into a company.

THE OUTPUTS

From atoms to impact - seven kinds of output.

1

Philosophical

A culture that says the future is built, not waited for.

2

Communal

A durable cross-disciplinary community and talent network.

3

Financial

Startups that raise, sponsorships, a self-sustaining platform.

4

Projects

Ambitious builds too big for a single course or lab.

5

Research

Papers, replications, lab-to-world translation.

6

Products

Prototypes that become real, sellable technology.

7

Startups & founders

IP, spinouts, founders, partnerships, open source.

WHY US, WHY NOW, WHY HERE

The right team, the right moment, the right place - on merit.

Why now

Physical AI is inflecting; software is commoditising; the builders exist but lack a space.

Why here

Aalto has the talent, the design DNA, and a proven record of student-led ventures going global.

Why us

Three shipping communities, 1,000+ people already engaged, partners already in — before a single euro of dedicated funding.

Our way of operating already produces meaningful output — and Finland's tech ecosystem structurally needs a hardware forge to renew its engineering heritage.

We know exactly who we're measuring against.

MIT

35+ makerspaces (Project Manus). Media Lab funds ~25% of a ~\$66–80M budget from ~34 corporate members. Fab labs began here.

ETH Zurich

Student Project House: 3,000+ students built their ideas since 2016; spinoffs like Yasai; pro-bono experts + fellowships.

Newlab

400+ member startups raised \$5.8B VC, \$2.3B exits, \$20B+ valuation — largely self-funded via corporate/government projects.

TAKOMO

The Nordic forge built to reach that tier - starting from a running community.

VISION

Make Aalto a serious global contender in hardware - a forge Europe looks to. (placeholder, not a good line)

1

Ecosystem

Strengthen the pipeline from research and technical projects to startups, open-source tech, and real-world impact.

2

Community

A shared home for ambitious engineers, researchers, founders, and builders across disciplines.

3

International branding

Establish Aalto among Europe's leading hubs for deep-tech, technical talent, and hands-on engineering.

THE PLAN

Here's exactly what we'll do this year.

- 1** **Now → Q4 2026**
Charter signed; shared space equipped and open; weekly build sessions, workshops, and demo nights running; membership + governance formalised.
- 2** **Q1–Q2 2027**
Monthly flagship programming; growing participation from students, builders, industry; 15+ partnerships across industry and the ecosystem.
- 3** **Q3 2027**
First flagship innovation program completed; outcomes measured and documented; expansion planning begins.
- 4** **Q4 2027**
Supported projects progress into startups (some VC-funded); TAKOMO recognised as a key platform connecting research, talent, and hardware in Finland.

TEAM & STRUCTURE

Built by people who already ship — with a clean, transparent backbone.

Founders & board

Thinkin' Rocks and HSA leadership.

Matias Korte

Yerzhan Zamashev

Artur Roos

Milana Begantsova

Oleksandr Osadchuk

Contact: matias.korte@thinkinrocks.com · +358 40 098 0609

Structure

- HSA ry owns 100% of Thinkin' Rocks Oy, which operates the TAKOMO space.
- ASA and AROC are founding communities with their own governance and budgets.
- Legal and bookkeeping are kept separate across entities; TAKOMO holds no assets.

THE ASK

What we need to light the forge.

For Aalto University

- A shared home to collaborate, host events, build infrastructure
- Strategic partnership & connection to research + industry
- Early-stage support to establish the forge

We're not seeking permanent support - we're building a platform that becomes self-sustaining.

For donors

- Fund [€X] to equip the forge + run year one
- Use of funds: build-out / tools / programming / ops
- Named recognition as a founding patron

Your impact: [N] builders supported, [N] projects, first spinouts.

For corporate sponsors

- Sponsorship tiers [€X / €Y / €Z]
- Recruiting pipeline, first look at IP & prototypes
- Branded challenges and on-site presence

Modelled on MIT Media Lab: pay for proximity to what's next.

Keep only the variant relevant to the room - always with a specific number and a use-of-funds split.

Let's build the future — from atoms to impact.

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matias.korte@thinkinrocks.com · +358 40 098 0609 · thinkinrocks.com